Week 2 Internship Report: Quantum Communication Theory

Noisy Quantum Channels and Information-Theoretic Limits

Connection to Week 1: Foundations and Framework

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Outline

1. Introduction: Connection to Week 1 Foundations
2. Mathematical Modeling of Fiber Noise
3. Exact Solution Methods
4. Holevo Capacity Calculation
5. Post-Channel CPTP Correction
6. Decoding/Inversion Ambiguity
7. Integrated Framework: Weeks 1 & 2
8. Progress Summary
9. Conclusion and Future Work

Week 1: Established Foundation

Week 1 Achievements

1. **Electromagnetic foundations**: Maxwell's equations, modal decomposition
2. **Quantum field theory**: Field quantization, coherent states
3. **Atom-field interaction**: Hamiltonian $H(t|\theta)$ with fiber parameters
4. **Cq channel theory**: Encoding, Holevo capacity
5. **Computational complexity**: NP-hardness via SJIP reduction

Week 2: Extensions to Noisy Channels

1. **Noise modeling**: GKSL master equation
2. **Exact dynamics**: Solution methods
3. **Noisy capacity**: Holevo capacity under noise
4. **Fundamental limits**: Quantum DPI
5. **Decoding ambiguity**: Formal characterization

From Ideal to Noisy Propagation

Week 1: Ideal Model

Coherent State: $|\alpha(0)\rangle \to |\alpha(L)\rangle$ with $\alpha(L) = \alpha(0)e^{i\beta L}e^{-\alpha L/2}$

- Noiseless, perfect information, deterministic

Week 2: Realistic Model with GKSL Master Equation

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -i[H,\rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right)$$

Fiber Noise Sources:

- Amplitude damping: γ_{loss}, Phase damping: γ_ϕ
- Non-unitary evolution, information loss, stochastic processes

Key Connection

Week 1's attenuation α becomes Week 2's damping: $\gamma_{\text{loss}} = \frac{\alpha v_g \ln(10)}{10}$
Ideal model = special case when $\gamma_{\text{loss}} = \gamma_\phi = 0$

GKSL Master Equation

Theorem 2.1 (GKSL Master Equation)

$$\frac{\mathrm{d}\rho_S}{\mathrm{d}t} = -i[H_S, \rho_S] + \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho_S\} \right)$$

Connection to Week 1 Parameters

$$\gamma_{\text{loss}} = \frac{\alpha v_g \ln(10)}{10}$$

$$\gamma_\phi = \frac{(\omega D_p \Delta L)^2 v_g}{2}$$

where α, D_p are Week 1 fiber parameters

Direct Diagonalization Method

Theorem 3.1 (Direct Diagonalization)

For d-dimensional system, Liouvillian $\mathbb{L}$ is $d^2 \times d^2$:

$$\vec{\rho}(t) = e^{\mathbb{L}t}\vec{\rho}(0)$$

Complexity

- Scales as $O(d^6)$
- Limited to small systems ($d \lesssim 10$)
- Exact up to numerical precision

Quantum Trajectory Method

For Non-Markovian Dynamics

$$\frac{\mathrm{d}}{\mathrm{d}t}\left|\psi(t)\right\rangle = -iH_{\text{eff}}(t)\left|\psi(t)\right\rangle + \sum_k \xi_k(t) L_k \left|\psi(t-\tau_k)\right\rangle$$

Advantages

- Handles large systems
- Captures exact noise correlations
- Parallelizable via Monte Carlo

From Ideal to Noisy Capacity

Week 1: Ideal Capacity

$$C(\theta) = \max_{p(x)} \chi(\{p(x), \rho(x|\theta)\})$$

For ideal channels

Week 2: Noisy Capacity

$$C_\chi(\mathcal{N}) = \max_{\{p(x), \rho_x\}} \chi(\{p(x), \mathcal{N}(\rho_x)\})$$

For noisy channel $\mathcal{N}$

Analytical Capacities for Specific Channels

Theorem 4.1 (Depolarizing Channel)

$$C_\chi^{dep} = 1 - H_2\left(\frac{p}{2}\right) - p\log_2 3$$

Theorem 4.2 (Fiber Noise Channel)

$$C_\chi^{fiber} \leq \min\left\{C_\chi^{AD}(\gamma), C_\chi^{PD}(\lambda)\right\}$$

For typical fiber parameters

$C_\chi \approx 0.7 - 0.9$ bits per photon

Quantum Data Processing Inequality

Theorem 5.1 (Quantum DPI)

For $\rho \to \sigma \to \tau$ with $\tau = \mathcal{C}(\sigma)$:

$$I(\rho; \tau) \leq I(\rho; \sigma)$$

Theorem 5.2 (Capacity Non-Increase)

For any $\mathcal{N}$ and CPTP $\mathcal{C}$:

$$C_\chi(\mathcal{C} \circ \mathcal{N}) \leq C_\chi(\mathcal{N})$$

Implications for Communication Strategy

Corollary 5.3 (Optimal Strategy Structure)

Optimal receiver must be:

$$\text{Measurement} \circ \mathcal{N}$$

not

$$\text{Measurement} \circ \mathcal{C} \circ \mathcal{N}$$

for any CPTP $\mathcal{C}$

Key Insight

- Post-channel correction cannot increase capacity
- Focus on optimal encoding and measurement
- Quantum error correction must be *before* channel

Channel Non-Injectivity

Theorem 6.1 (Non-Injectivity of Noisy Channels)

For any non-unitary $\mathcal{N}$, $\exists \rho_1 \neq \rho_2$ such that:

$$\mathcal{N}(\rho_1) = \mathcal{N}(\rho_2)$$

Definition 6.1 (Ambiguity Set)

$$\mathcal{A}(\sigma) = \{\rho' : \mathcal{N}(\rho') = \sigma\}$$

Capacity-Ambiguity Trade-off

Theorem 6.2 (Capacity-Ambiguity Relation)

$$C_\chi(\mathcal{N}) \leq \log_2\left(\frac{1}{\mathcal{A}_{\min}}\right)$$

where $\mathcal{A}_{\min}$ is minimum achievable ambiguity

Connection to Week 1

- Week 1: NP-hardness via SJIP reduction
- Week 2: Additional ambiguity from non-injectivity
- Combined: Even harder decoding problem

Theoretical Progression

Stage	Week 1: Foundations	Week 2: Extensions
Physical	Maxwell, quantization	GKSL, exact solutions
Information	Holevo capacity	Noisy capacities, DPI
Computational	SJIP hardness	Ambiguity characterization
Parameters	$\theta = \{L, \alpha, \beta_2, \ldots\}$	$\gamma = \{\gamma_{\text{loss}}, \gamma_\phi, \ldots\}$

Key Theorem Connections

Week 1 Theorems	Week 2 Extensions
Field quantization: $\hat{A} = \sum_m \sqrt{\frac{\hbar\omega_m}{2\epsilon_0 V_m}}(\hat{a}_m f_m + \text{h.c.})$	GKSL: $\frac{\mathrm{d}\rho}{\mathrm{d}t} = -i[H, \rho] + \sum_k \gamma_k (L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$
Holevo bound: $I(X : Y) \leq \chi(\{p(x), \rho(x\|\theta)\})$	Capacity non-increase: $C_\chi(\mathcal{C} \circ \mathcal{N}) \leq C_\chi(\mathcal{N})$
SJIP NP-hardness: Subset-Sum $\leq_p$ SJIP	Non-injectivity: $\exists \rho_1 \neq \rho_2$ s.t. $\mathcal{N}(\rho_1) = \mathcal{N}(\rho_2)$

Week 1 Achievements

1. Established complete electromagnetic foundation
2. Derived quantum field quantization in fibers
3. Developed Cq channel theory with parameter dependence
4. Proved NP-hardness of optimal decoding via SJIP
5. Provided explicit formulas for system design

Week 2 Extensions

1. Extended physical modeling to include noise via GKSL
2. Developed exact solution methods for noisy dynamics
3. Calculated Holevo capacities for various noisy channels
4. Proved fundamental limits via Quantum DPI
5. Characterized decoding ambiguity under noise
6. Integrated all results with Week 1 foundations

Key Integrated Insights

1. **Physical-Information Connection:**

$$\gamma_{\text{loss}} = \frac{\alpha v_g \ln(10)}{10}, \quad \gamma_\phi = \frac{(\omega D_p \Delta L)^2 v_g}{2}$$

2. **Fundamental Limits:**

$$I(X:Y) \leq \chi(\{p(x), \rho(x|\theta)\})$$
$$C_\chi(\mathcal{C} \circ \mathcal{N}) \leq C_\chi(\mathcal{N})$$

3. **Computational Reality:** NP-hardness + ambiguity
4. **Practical Guidance:** Explicit design formulas

Future Directions

Building on Both Weeks

1. **Experimental validation** with fiber-optic testbeds
2. **Algorithm development** respecting complexity constraints
3. **System optimization** using parameter-dependent formulas
4. **Extension** to channels with memory, nonlinear effects

Optimization Problem

$$\max_{\theta} C(\theta) = \max_{\theta} \max_{p(x)} \chi(\{p(x), \rho(x|\theta)\})$$

Final Remarks

Complete Theoretical Picture

- **Connects** physical reality to information limits
- **Bridges** ideal theory with practical considerations
- **Links** information limits with computational limits
- **Provides** explicit guidance for system design

Integrated framework offers both theoretical insights and practical engineering value

Appendix: Mathematical Connections

From Week 1 to Week 2 Propagation

Week 1 ideal:

$$|\alpha(0)\rangle \to |\alpha(L)\rangle, \quad \alpha(L) = \alpha(0)e^{i\beta L}e^{-\alpha L/2}$$

Week 2 noisy (amplitude damping):

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\eta(t)} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{\eta(t)} \\ 0 & 0 \end{pmatrix}$$

with $\eta(t) = 1 - e^{-\gamma_{\text{loss}} t}$

Appendix: Parameter Dependencies

Across Weeks

$$\text{Week 1: } g(\theta) = d_{eg}\sqrt{\frac{\omega_0^3}{2\hbar\epsilon_0 V_{\text{eff}}(\theta)}}|f_0(r_{\text{atom}}|\theta)|$$

$$\text{Week 2: } \gamma_{\text{loss}} = \frac{\alpha v_g \ln(10)}{10}$$

Connection Insight

Smaller V_{eff} (Week 1) increases coupling but also affects loss rates (Week 2)

Thank You

Questions?